TABLE 4-3
A *machine code* program for adding 1234 and 4321. This is the lowest level of programming: direct manipulation of the digital electronics. (The right column is a continuation of the left column).

```
10111001        00000000
11010010        10100001
00000100        00000000
10001001        00000000
00001110        10001011
00000000        00011110
00000000        00000010
10111001        00000000
11100001        00000011
00010000        11000011
10001001        10100011
00001110        00000100
00000010        00000000
```

machine code, since there is a one-to-one correspondence between the program commands and the action taken in the microprocessor. For example: ADD AX, BX translates to: 00000011 11000011. A program called an **assembler** is used to convert the assembly code in Table 4-4 (called the **source code**) into the patterns of ones and zeros shown in Table 4-3 (called the **object code** or **executable code**). This executable code can be directly run on the microprocessor. Obviously, assembly programming requires an extensive understanding of the internal construction of the particular microprocessor you intend to use.

TABLE 4-4
An *assembly* program for adding 1234 and 4321. An *assembler* is a program that converts an assembly program into machine code.

```
MOV CX,1234         ;store 1234 in register CX, and then
MOV DS:[0],CX       ;transfer it to memory location DS:[0]

MOV CX,4321         ;store 4321 in register CX, and then
MOV DS:[2],CX       ;transfer it to memory location DS:[0]

MOV AX,DS:[0]       ;move variables stored in memory at
MOV BX,DS:[2]       ;DS:[0] and DS:[2] into AX & BX

ADD AX,BX           ;add AX and BX, store sum in AX

MOV DS:[4],AX       ;move the sum into memory at DS:[4]
```

Assembly programming involves the direct manipulation of the digital electronics: registers, memory locations, status bits, etc. The next level of sophistication can manipulate *abstract* variables without any reference to the particular hardware. These are called **compiled** or **high-level** languages. A dozen or so are in common use, such as: C, BASIC, FORTRAN, PASCAL, APL, COBOL, LISP, etc. Table 4-5 shows a BASIC program for adding 1234 and 4321. The programmer only knows about the variables A, B, and C, and nothing about the hardware.

TABLE 4-5
A *BASIC* program for adding 1234 and 4321. A *compiler* is a program that converts this type of high-level source code into machine code.

```
100  A = 1234
110  B = 4321
120  C = A+B
130  END
```